

TERRA: an accessible, HPC-integrated pipeline for downstream transposable-element family analysis and CRISPR reagent design

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July 16, 2026

Abstract

Transposable elements (TEs) make up a large fraction of mammalian genomes and are increasingly recognised as functional contributors to genome regulation in development and disease [1]. Mature tools discover TE families from raw assemblies [2] and quantify repeat expression [3–5], but what happens once a family is named is under-served: deciding which copies matter and turning that into reagents a bench scientist can order. We present **TERRA**, a modular open-source pipeline that takes a TE family name and a reference assembly and returns two reagent deliverables, **family- and locus-specific primers** and **allele-aware CRISPRa/i guide RNAs** scored against the full copy landscape on a coverage-versus-off-target frontier, together with the evidence needed to interpret them. Because averaging across a family mixes a minority of expressed, intact copies with a majority of inert relics, TERRA clusters copies without alignment, separates the *expressed* from the *common* consensus, and layers expression, motif, phylogenetic and structural evidence on each sub-group, emitting figures and tables at every stage: the cluster embedding, per-cluster expression and differential-expression tests, MAFFT alignments and consensus sequences, JASPAR motif enrichment, and a molecular-clock phylogeny that nominates a master element. On the prototype family *L1Md_T* in mm10, 23,639 annotated loci reduce to 5 clusters, 7,490 carry SQuIRE expression [3], 9388 copies retain ORF integrity, and the pipeline nominates a master-element candidate (**chrX:40304675**) and 400 guides of which 17 are non-dominated. The same run executes unchanged on a laptop or on LSF/Slurm clusters through Nextflow DSL2 [6, 7]. TERRA is open source.

1 Introduction

Interspersed repeats derived from TEs account for roughly half of the human genome [8] and a comparable or larger share of many other eukaryotic genomes [1], where they shape genome organisation, supply regulatory sequence, and influence host gene expression in development and disease [9, 10]. The natural unit of analysis is the *family*, which groups copies descended from a common ancestral element.

The bioinformatics of TEs has historically concentrated on two upstream problems: building libraries of family models *de novo* from an assembly, and annotating a genome against them. Tools such as RepeatModeler2 [2] and the RepeatMasker/Dfam ecosystem [11] address both well. A second, well-populated group quantifies repeat expression, including SQuIRE [3], TEtranscripts/TElocal [4], Telescope [5], scTE [12] and ERVmap [13].

The under-served workflow begins once a family has been named and quantified, and two problems sit at its centre. First, which copies matter: a named family bundles a few expressed, intact insertions with many truncated, inert relics, so any family-level analysis dilutes a signal

carried by a minority of loci. Resolving the family into sub-groups and asking which are expressed is what makes the result interpretable, and it produces the evidence a biologist reads directly (cluster embedding, per-cluster expression, alignments, consensus sequences, motif enrichment). Second, the reagents: most TE projects end in an experiment, and repetitiveness makes the necessary primers and guides hard to design, since a guide inevitably matches many copies and must trade coverage of the family against off-target hits. TERRA treats primers and guides as first-class outputs, reports the guide trade-off as a Pareto frontier rather than one score, and returns both the reagents and the evidence behind them as figures and tables rather than leaving the user to assemble them.

TERRA applies the philosophy of integrated analysis platforms [6, 7, 14, 15], namely hide the infrastructure, keep every step reproducible, and run anywhere. It is not a TE-discovery tool and does not compete with RepeatModeler2 [2]: it begins with a defined family and assembly and automates the characterisation and reagent design that follow, with first-class support for the LSF and Slurm schedulers that dominate academic HPC.

Figure 1 summarises the path this paper follows end to end on a single prototype family.

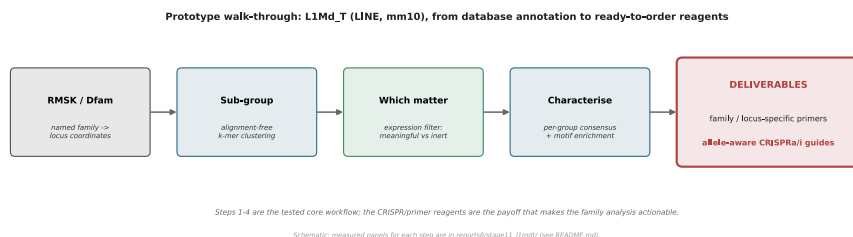


Figure 1: The TERRA path, from database annotation to ordered reagents, illustrated for the prototype family used throughout this paper. RepeatMasker/Dfam loci are sub-grouped without alignment, filtered by locus-level expression to separate copies that matter from inert relics, characterised by consensus and motif content, and converted into the two deliverables: family/locus-specific primers and allele-aware CRISPRa/i guides. This panel is a schematic of the workflow; every measured result it refers to is reported in Section 2.

2 Results: L1Md_T from annotation to reagents

We use one family, *L1Md_T*, as a prototype throughout: a young, active mouse LINE-1 subfamily that is large, contains both intact and heavily truncated copies, and is expressed in early development, so every stage is exercised. All results come from a single *mm10* run with locus-level expression from SQuIRE [3] across six early-development stages (oocyte, pronucleus, 2-, 4-, 8-cell, morula). A SINE (*B1_Mus2*) and an LTR (*IAPLTR1_Mm*) family, showing the workflow generalises across TE classes, are in Section 7 and the Supplementary Files.

2.1 Starting point: the annotated family

TERRA takes the family name and assembly and retrieves loci and per-locus divergence from the RepeatMasker track [11], then extracts sequence from the reference. For *L1Md_T* in *mm10* this yields 23,639 annotated loci. This is the number a biologist starts with, and on its own it is not actionable: it says nothing about which of those 23,639 insertions are intact or worth targeting.

2.2 Resolving copies rather than lumping them

Clustering is alignment-free. Each locus sequence is encoded as a k -mer count vector (default $k=6$, giving a $4^6 = 4096$ -dimensional sparse vector built with scikit-learn [16]), reduced to 50 components by truncated SVD, embedded in two dimensions with UMAP [17], and partitioned with HDBSCAN [18]. This avoids a family-wide multiple alignment and scales to tens of thousands of loci. For *L1Md_T* the parameter search (Section 4) converged to a divisor of $N/5$ with UMAP `n_neighbors` = 50, $k = 5$, minimum cluster size 2000 and `min_samples` = 25, partitioning the family into 5 clusters with 7.2% of loci left in HDBSCAN’s noise class.

These parameters were not taken on faith. The pipeline’s heuristic minimum cluster size $\lfloor N/5 \rfloor$ demands 4,727 loci per cluster for *L1Md_T*, which only 2 groups can satisfy, forcing 53.8% of the family into the noise class; that is a property of the threshold, not the family. Searching k , `n_neighbors`, `min_cluster_size` and `min_samples` (288 configurations, `run_cluster_search.py`), 44 recovered 5–6 clusters, and the selected one cuts the unassigned fraction from 53.8% to 7.2% (Figure 6). The remaining noise class is kept as a labelled group but excluded from consensus building (Section 4), so unassigned loci cannot contaminate a consensus that reagents target.

The payoff is shown in Figure 4: a family-level mean is dominated by the inert majority, burying a real effect in the expressed, intact minority. Resolving copies and intersecting them with expression separates these populations, and the consensus built from expressed copies (the *expressed* consensus) is what reagents should target, which is not in general the consensus built from all annotated copies.

Of the 23,639 loci, 7,490 (roughly a third) carry SQuIRE expression entries, summarised per cluster (units in Section 4). Pairwise Wilcoxon rank-sum tests with Benjamini–Hochberg correction find 54 of 60 cluster-pair by stage comparisons significant at $q < 0.05$ across the 6 stages and 5 clusters. Expression is uneven: it concentrates in a minority of clusters and peaks at the 8-cell stage while the rest are near-silent (Figure 2b), exactly the separation a family-level average would remove. Figure 2 shows the partition, the per-cluster expression profile and the divergence distribution.

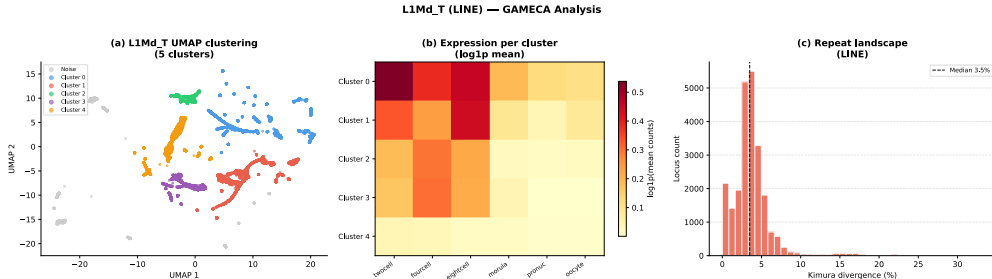


Figure 2: *L1Md_T* (mm10) analysed end to end in a single TERRA run. (a) UMAP embedding of k -mer-encoded loci coloured by HDBSCAN cluster. (b) Per-cluster mean SQuIRE expression ($\log_{1p}$ mean) across the six developmental stages, computed from the 7,490 loci carrying expression entries. (c) CpG-corrected Kimura divergence histogram, median 3.5%.

2.3 Age, intactness, and the master element

Copies are aligned with MAFFT [21], a majority-rule consensus is built, and each copy’s Kimura two-parameter divergence [22] from that consensus is converted to a molecular-clock age. ORF

integrity separates intact from degraded copies, and a master-element score combining intactness and divergence nominates the youngest, most intact copies, which are the ones most likely to still be capable of mobilisation. For *L1Md_T*, 9388 of 23,639 copies retain ORF integrity, the median copy age is 5.41 Myr, and the top master-element candidate is chrX:40304675 at 7.23% divergence from the consensus (Figures 8 and 5).

2.4 Deliverable 1: allele-aware CRISPRa/i guides

Designing CRISPR reagents against a repetitive family is the problem most tools leave to the user. A guide has no single on-target site: it matches some number of family copies and some elsewhere, so the design question is bi-objective (maximise on-family coverage, minimise off-target hits) and cannot be reduced to one score without hiding the trade-off. TERRA enumerates PAM-aware candidate guides from the consensus and per-copy sequence, scores each against the *full* copy landscape (and an optional background FASTA) with seed-anchored, mismatch-tolerant matching, and reports the coverage-versus-off-target Pareto frontier. For *L1Md_T*, 400 guides scored against all 23,639 copies yield 17 non-dominated design choices (Figure 3), the best being AGGGAGGGAAGGAGGGAGGG; selecting from the frontier is choosing a documented position on the trade-off rather than trusting an opaque ranking. The same output supports CRISPRa and CRISPRi, which depend on the guide-to-copy mapping rather than the effector (Figure 9).

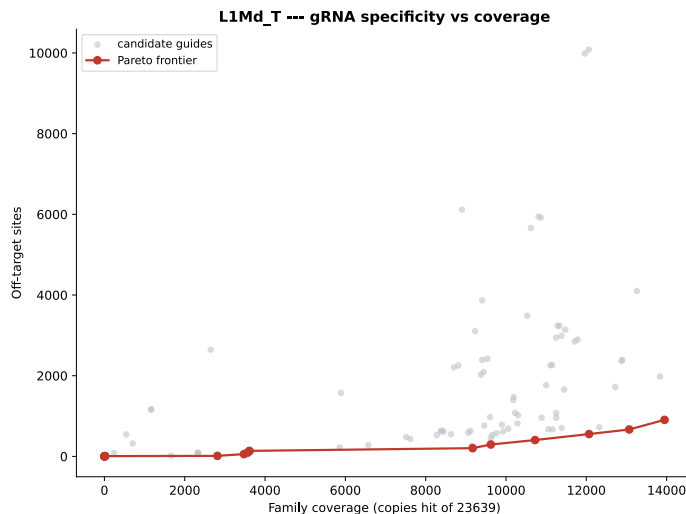


Figure 3: Measured coverage-versus-off-target frontier for *L1Md_T*. 400 candidate guides scored against 23,639 copies; 17 guides are non-dominated. Each frontier point is a defensible design choice at a stated coverage and off-target count.

2.5 Deliverable 2: family- and locus-specific primers

Primers are proposed from frequent k -mers in the (expressed) consensus and each candidate is checked for specificity genome-wide across all chromosomes. Because the specificity check is the most I/O-intensive step in the pipeline, it motivates the genome-cache engineering described in Section 5. Primer candidates are emitted as a table with their genome-wide match counts, and are cross-referenced against the per-locus expression so that a user can select primers that report on the copies that are actually transcribed rather than on the family as annotated.

2.6 Additional copy-resolved evidence

Three further modules were exercised on *L1Md_T* and are reported here because they bear directly on reagent design. 3' transduction detection groups copies sharing downstream-tail *k*-mers absent from the consensus, alongside poly-A signal detection, and resolves 30 candidate lineages with 8227 copies carrying an AATAAA signal (Figure 10). Antisense/bidirectional promoter scanning of each 5' region on both strands flags 13246 of 23,639 copies as bidirectional (Figure 11), which is relevant because a CRISPRa guide placed in a bidirectional promoter may drive transcription in both directions. A further set of modules (CTCF/TAD proximity, epigenetic overlay, orthologous-insertion calling, multi-assembly liftover, ColabFold structure prediction, LTR structural classification, and automatic subfamily resolution) place the same resolved copies in regulatory, evolutionary and structural context, and are described in Section 6.

3 How TERRA relates to existing tools

TERRA's neighbours fall into three groups (Table 1), and it consumes the output of the first two rather than competing with them. *Discovery and annotation*: RepeatModeler2 [2] and the RepeatMasker/Dfam ecosystem [11] produce the locus coordinates TERRA starts from. *Expression quantification*: SQuIRE [3], TEtranscripts/TElocal [4], Telescope [5], scTE [12], ERVmap [13] and TE-Seq [23] assign reads to repeats; TERRA consumes their per-locus counts (SQuIRE here) and does not requantify. These answer "how much is expressed"; TERRA addresses which copies to group, what their consensus is, and what reagent targets them. *General platforms*: Galaxy [14], GenePattern [15] and nf-core/Nextflow [6, 7] are portable substrates but ship no ready-made TE-family workflow; TERRA builds on them, shipping a Nextflow DSL2 implementation importable as a subworkflow. Table 1 compares *scope and access* from documented capabilities; no cross-tool runtime is claimed.

Tool	TE-family focus	Copy-resolved character.	Reagent design	Graphical interface	Built-in LSF/Slurm	Reprod. environ.
TERRA (this work)	✓	✓	✓	✓	✓	✓
RepeatModeler2 [2]	✓	×	×	×	×	○
RepeatMasker/Dfam [11]	✓	×	×	×	×	○
SQuIRE [3]	✓	○	×	×	×	○
TEtranscripts/TElocal [4]	✓	○	×	×	×	○
Telescope [5]	✓	○	×	×	×	○
scTE [12]	✓	○	×	×	×	○
ERVmap [13]	✓	○	×	×	×	○
TE-Seq [23]	✓	○	×	×	○	○
Galaxy [14]	×	○	×	✓	○	✓
GenePattern [15]	×	○	×	✓	○	✓
nf-core/Nextflow [7]	×	○	×	×	✓	✓

Table 1: Capability comparison on documented features. ✓ built-in; ○ partial, or possible but the user must assemble or configure it; × not provided. *Copy-resolved characterisation*: partitioning a named family into sub-groups and building per-group consensus, as distinct from quantifying expression per locus. *Reagent design*: primers and/or allele-aware CRISPR guides emitted for the family. Expression quantifiers are marked ○ for copy-resolved characterisation because they report per-locus or per-family counts without grouping copies or building consensus. No cross-tool runtime is claimed.

4 Validation and reporting conventions

Expression is consumed as per-locus counts from an external quantifier (SQuIRE [3] throughout); TERRA does not requantify. Per-cluster values are means of $\log_{1p}$ per-locus counts over the loci in that cluster carrying an entry; loci with no entry are excluded rather than treated as zero, which is why 7,490 of 23,639 loci enter the analysis. Motif results are counts of JASPAR matches [24] per locus intersected with BEDTools [25]; enrichment is a per-cluster Fisher exact test on loci-carrying-a-match, so a locus with several matches for one motif contributes once.

HDBSCAN labels loci it cannot confidently place as -1 . These are retained and reported as a labelled group but are excluded from per-cluster consensus building, so a locus never contributes to a consensus it was not assigned to. The adopted configuration leaves 7.2% of loci unassigned.

The partition is a stratification of copy state, not a subfamily assignment; three measurements bound what it can be claimed to be, reported with their negative results (Appendix A). *Parameter dependence*: across 9 configurations of k and `n_neighbors` the cluster count ranged 2–6 with median ARI 0.40 against the adopted partition, better than the superseded default (ARI 0.21, 2 clusters, 53.8% unassigned) but short of a subfamily claim. *Reproducibility*: clustering is deterministic within an environment (ARI 1.000 on re-run), but UMAP runs with `n_jobs = -1`, so counts can shift with core count; every figure uses the stored labels and embedding. *Consensus cross-check*: projecting each sampled copy onto every per-cluster consensus with `mafft -addfragments -keeplength`, only 36.1% are nearest their own cluster’s consensus (chance 20.0%) and the median copy is marginally closer to another’s (0.035 vs 0.030), so we do not call the clusters subfamilies. What they capture is degradation state: ORF integrity runs from 91.2% intact in the most intact cluster to 0.1% in the least, and expression concentrates there, which is the axis the workflow needs.

The core workflow together with the phylogenetics, gRNA, transduction and antisense modules is the source of every number reported above; the further analyses of Section 6 extend the same resolved family into regulatory, evolutionary and structural context.

5 Design and implementation

TERRA is organised in three layers (Figure 14). A Tauri/React desktop application gives point-and-click access and performs no analysis itself; it launches a bundled Python sidecar over newline-delimited JSON, so jobs stream progress and can be cancelled. The backend is a driver (`query.py`) orchestrating the core stages, the standalone `run_*.py` modules and an SSH client: locally it fetches sequence from the UCSC API [26] when no FASTA is present, and on HPC it stages code, submits a batch job and retrieves results. Every external resource (RMSK/Dfam, UCSC [26], JASPAR [24], mygene.info [27]) sits behind a thin adapter, so adding an assembly or source is a localised change.

The pipeline is a directed sequence of stages (Table 2). Each stage reads the previous stage’s tabular output and writes a checkpoint, so a run can be resumed or a single stage re-executed. Figure 15 gives the end-to-end view.

The dominant cost in whole-family primer design is reading the reference genome, since a naive implementation re-reads the multi-gigabyte FASTA for every candidate search. TERRA parses the reference once into an in-memory genome cache, reused for sequence extraction and all primer searches, and persists it so later runs skip parsing. A small number of batch operations (GC content, length tabulation, k -mer matrix construction) dominate the per-sequence arithmetic and have Cython [29] implementations with a pure-Python fallback.

The driver can run the copy-resolved modules as a sequential subprocess loop, or delegate

them to a Nextflow DSL2 layer [6] that runs the same unmodified Python modules as parallel, individually-resourced, resumable tasks (Figure 16). No scientific code is duplicated between the two.

6 Additional analysis modules

TERRA ships further modules that extend the resolved family into its regulatory, evolutionary and structural context (Figure 17); each runs as an independent task on the same per-copy tables and emits its own figures, values and provenance manifest. *Regulatory*: CTCF-site/TAD-boundary proximity and an epigenetic/regulatory-track overlay separate silenced from active insertions, complementing the expression stratification of Section 2.2. *Evolutionary*: orthologous-insertion calling and multi-assembly liftover (hg38, T2T-CHM13, mm10, mm39) place each copy across assemblies and species, and automatic subfamily resolution collapses the per-cluster consensus into a dendrogram. *Structural*: an LTR structural classifier labels each element (full-length / solo-LTR / internal-only / truncated) and ColabFold folds the consensus ORFs. Each runs on any family the core workflow accepts and writes the same output as the core stages.

7 Generality across TE classes

To show the workflow is not LINE-specific, we ran the identical pipeline on a SINE and an LTR family in mm10 with the same six-stage SQuIRE dataset. *B1_Mus2* (SINE) has 70,685 loci (21,407 expressed, median divergence 7.5%) and *IAPLTR1_Mm* (LTR) has 1,485 loci (1,426 expressed, 1.6%). The divergence medians order the families by age, *IAPLTR1_Mm* youngest, *L1Md_T* intermediate (3.5%) and *B1_Mus2* oldest, consistent with their retrotransposition histories [1]. Full per-family reports (clustering, expression, alignment, consensus, motif, primer and guide outputs) are in the Supplementary Files.

8 Discussion

TERRA fills a specific gap: the accessible, reproducible, cluster-ready path from a named TE family to reagents that can be ordered. It complements rather than replaces discovery tools such as RepeatModeler2 [2], expression quantifiers such as SQuIRE [3] and TEtranscripts [4], and general platforms such as Galaxy [14] and nf-core [7].

Its contributions are threefold: conceptually, resolving a family into sub-groups and intersecting them with locus-level expression lets a signal carried by a minority of copies survive, and motivates the expressed-versus-common consensus distinction (Section 2.2); practically, treating primers and allele-aware CRISPRa/i guides as the deliverable and reporting the guide trade-off as a frontier rather than one score (Section 2.4); and in engineering, the in-memory genome cache, compiled hot paths and automatic LSF/Slurm support make whole-family analysis practical from laptop to cluster.

Limitations bound the system. Expression quality and its multi-mapping policy are inherited from the upstream quantifier, and family resolution depends on external services (RMSK/Dfam, UCSC, JASPAR, mygene.info), so a run is only as reproducible as the database releases it draws on; supported assemblies are currently the common human and mouse builds. The *k*-mer clustering, though fast and alignment-free, does not yield stable sub-family boundaries at the default minimum

cluster size (agreement near chance, median ARI 0.40), so we report the partition as a device for separating expressed from inert copies, not a subfamily assignment; and the guide frontier reports sequence-level predictions that require experimental confirmation.

Natural extensions include per-locus epigenomic context (WGBS, ATAC-seq) to separate silenced from derepressed copies, correlating per-cluster activity with nearby gene expression, somatic- insertion calling from user BAMs, broader assembly support, and negative-binomial modelling of count overdispersion [30, 31].

Code availability

TERRA is open source at <https://github.com/anmol-dash/te-scraper-and-analysis>; the pipeline scripts there generate every figure and measured value cited here.

A Supplementary figures and tables

The figures and table below support the results and validation reported in the main text. They are placed here to keep the main narrative on the prototype walk-through and the reagent deliverables; each is referenced from the section it belongs to.

Stage	Function	Key methods / tools
Prep	Loci, sequences, divergence	RMSK/Dfam [11]; FASTA or UCSC [26]; Kimura milliDiv
Clustering	Sub-group the copies	k -mers $\rightarrow$ SVD $\rightarrow$ UMAP [17] $\rightarrow$ HDBSCAN [18]
Alignment	Consensus per group	MAFFT [21] + CIALign [28]
Motif	Enriched TFBS	JASPAR [24] $\cap$ loci (BEDTools [25]) + Fisher
GO	Gene/ontology context	mygene.info [27]
Expression	Per-group profiles + DE	per-locus counts (SQuIRE [3]); Wilcoxon + BH
Primers	Family/locus primers	k -mer candidates + genome-wide specificity
gRNA	Allele-aware CRISPRa/i	PAM-aware candidates + seed-anchored off-target scoring

Table 2: The TERRA core stages. Each stage consumes a tabular checkpoint and emits one, allowing partial re-runs. The gRNA and primer stages produce the deliverables of Section 2.4.

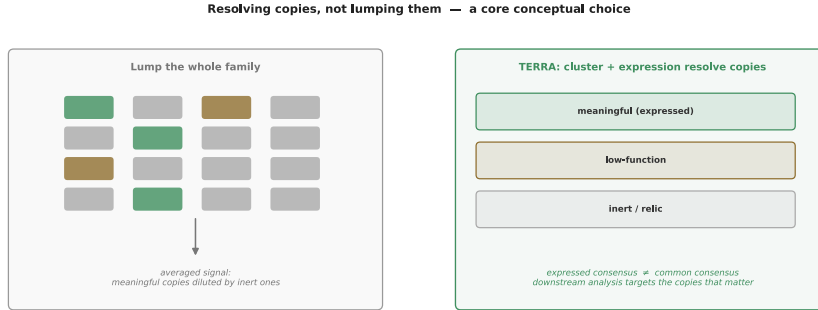


Figure 4: Resolving copies instead of lumping them. Left: a family-level summary averages over meaningful, low-function and inert loci, so signal carried by a minority of copies is diluted. Right: TERRA partitions copies by sequence and intersects them with locus-level expression, separating the populations and distinguishing the expressed consensus from the common consensus. This panel is a schematic of the concept; the measured partition for *L1Md_T* is in Figure 2.

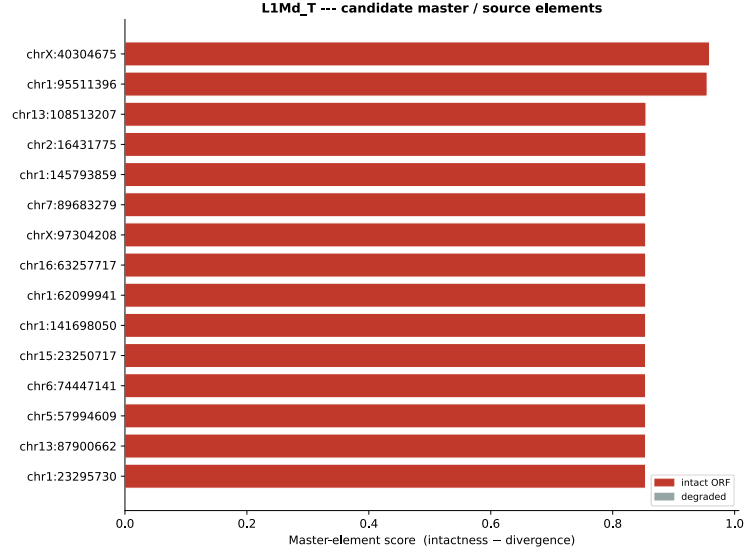


Figure 5: Master-element nomination for *L1Md_T*. Copies are scored on ORF integrity against divergence from the consensus; 9388 copies are intact and the top-ranked candidate is **chrX:40304675**.

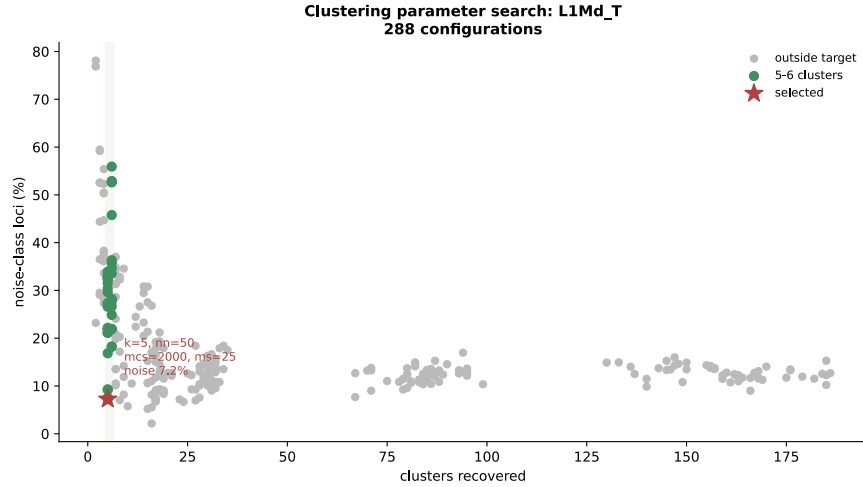


Figure 6: Clustering parameter search for *L1Md_T*. Each point is one of 288 configurations of k , $n_neighbors$, $min_cluster_size$ and $min_samples$, plotted as the number of clusters recovered against the fraction of loci left unassigned. The pipeline’s default minimum cluster size ($\lfloor N/5 \rfloor = 4,727$) admits only 2 clusters and leaves 53.8% of the family unassigned; the selected configuration (star) recovers 5 clusters with 7.2% unassigned.

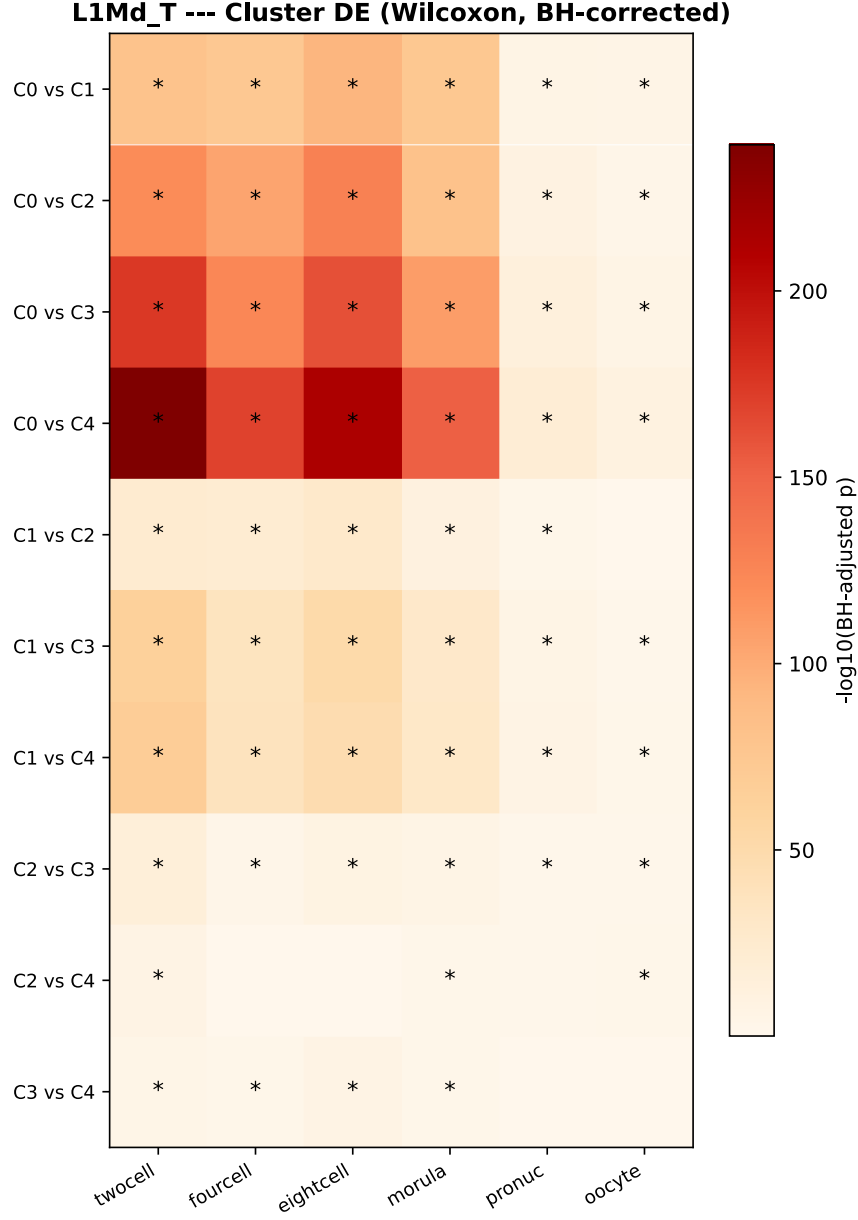


Figure 7: Differential expression between *L1Md_T* clusters. Heatmap of $-\log_{10}(q)$ from pairwise Wilcoxon rank-sum tests with Benjamini–Hochberg correction; 54 of 60 cluster-pair by stage comparisons reach $q < 0.05$. Cluster-resolved testing recovers stage-specific differences that a family-level summary averages away (Section 2.2).

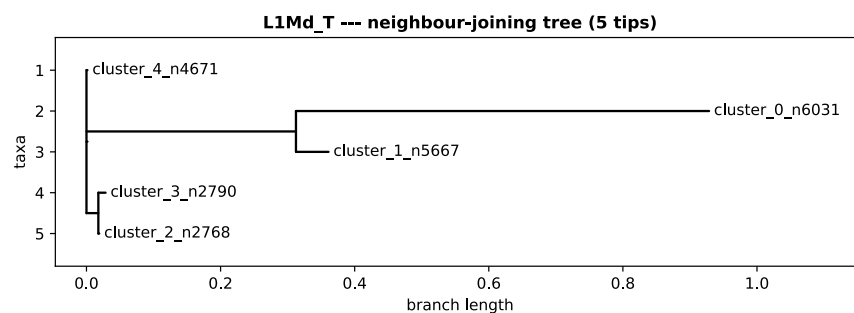


Figure 8: Subfamily phylogeny for *L1Md_T*. Neighbour-joining tree relating cluster consensus and representative copies, built from the MAFFT alignment.

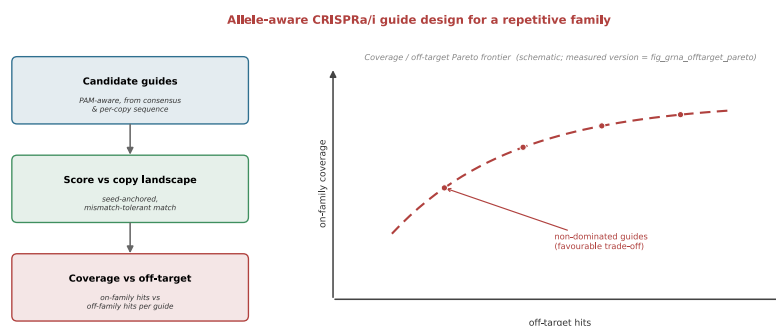


Figure 9: Allele-aware CRISPRa/i guide design for a repetitive family. Candidate guides are enumerated PAM-aware from the consensus and per-copy sequence, scored against the full copy landscape with seed-anchored, mismatch-tolerant matching, and reported as a coverage-versus-off-target trade-off. This panel is a schematic of the logic; the measured frontier for *L1Md_T* is Figure 3.

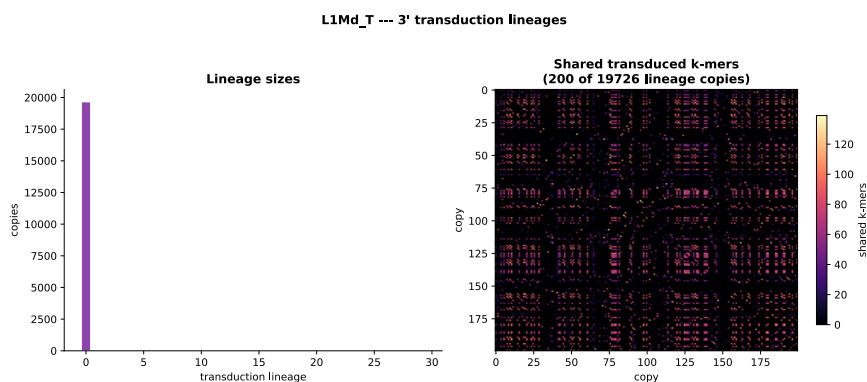


Figure 10: 3' transduction lineages for *L1Md_T*: 30 candidate lineages grouped by shared downstream-tail *k*-mers absent from the family consensus; 8227 copies carry an AATAAA poly-A signal.

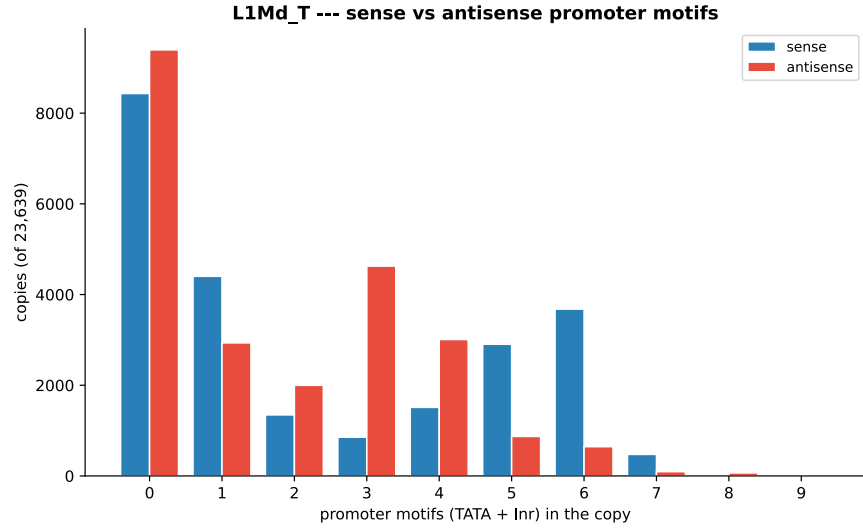


Figure 11: Antisense/bidirectional promoter content for *L1Md_T*: core promoter elements scanned on both strands of each 5' region; 13246 of 23,639 copies are flagged bidirectional.

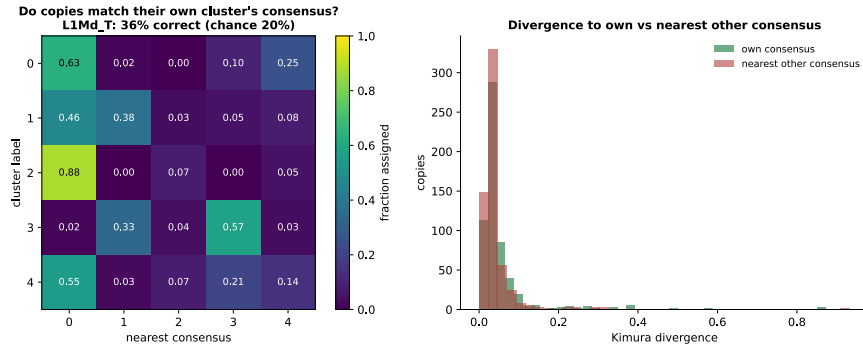


Figure 12: Cross-check of the clusters against the consensus alignments for *L1Md_T*. **Left:** fraction of each cluster's copies whose nearest consensus is each cluster's, after projecting every copy onto every consensus with `mafft -addfragments -keeplength`. A subfamily assignment would put the mass on the diagonal; the observed 36.1% (chance 20.0%) does not. **Right:** Kimura divergence of each copy to its own cluster's consensus versus to the nearest other consensus; the distributions overlap, with the median copy marginally closer to another cluster's consensus.

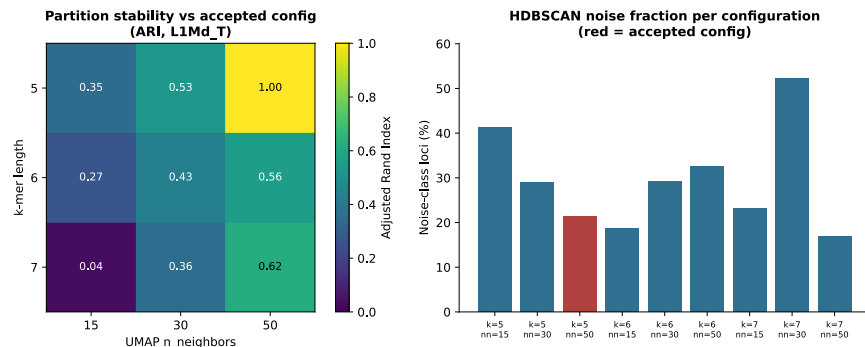


Figure 13: Clustering and noise-class validation for *L1Md_T*. **Left:** Adjusted Rand Index of each k and $n_neighbors$ configuration against the configuration the automatic search accepted. Values near 1 would indicate the partition is independent of the parameter choice; the observed median of 0.40 indicates it is not, which is why these sub-family boundaries are not interpreted as biological. **Right:** HDBSCAN noise-class fraction per configuration, with the accepted configuration highlighted. Noise-class loci are retained as a labelled group and excluded from consensus building.

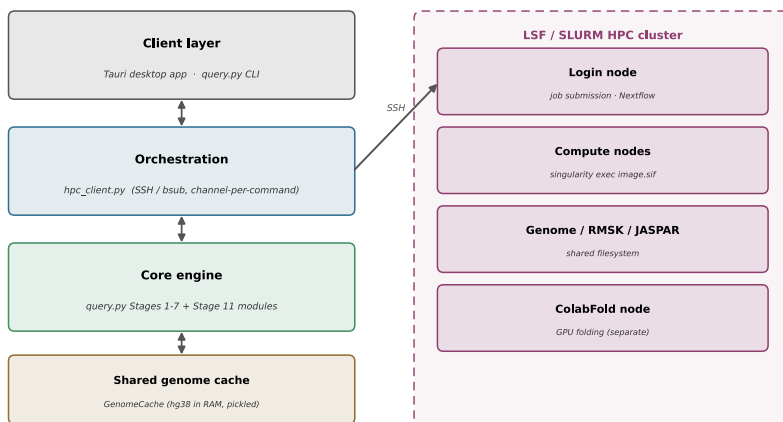


Figure 14: TERRA's three-layer architecture. A Tauri/React desktop shell drives a Python analysis sidecar over a streaming NDJSON protocol; the backend runs locally or stages and submits jobs to an LSF/Slurm cluster. Coordinate, sequence, motif and annotation data are pulled from RMSK/Dfam, UCSC, JASPAR and mygene.info through uniform adapters.

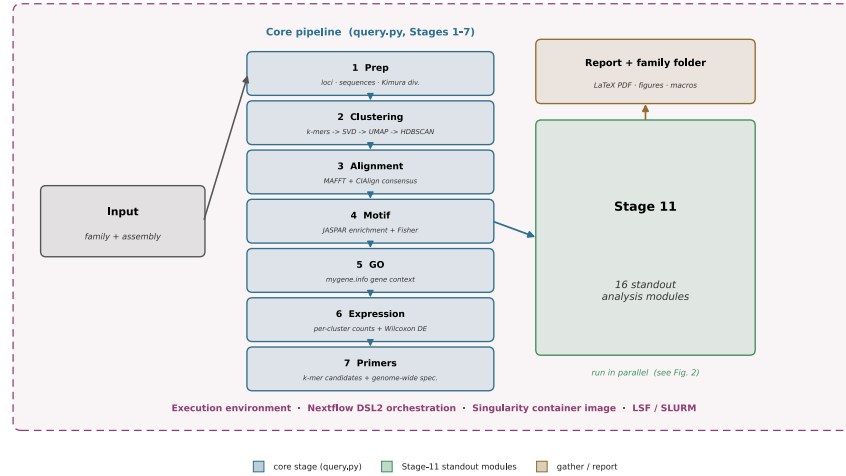


Figure 15: TERRA pipeline flowchart. A family/assembly request flows through the core stages and into the copy-resolved modules, which run in parallel; outputs are gathered into a per-family folder and report. The run executes inside the Nextflow DSL2 and Singularity environment on an LSF or Slurm scheduler.

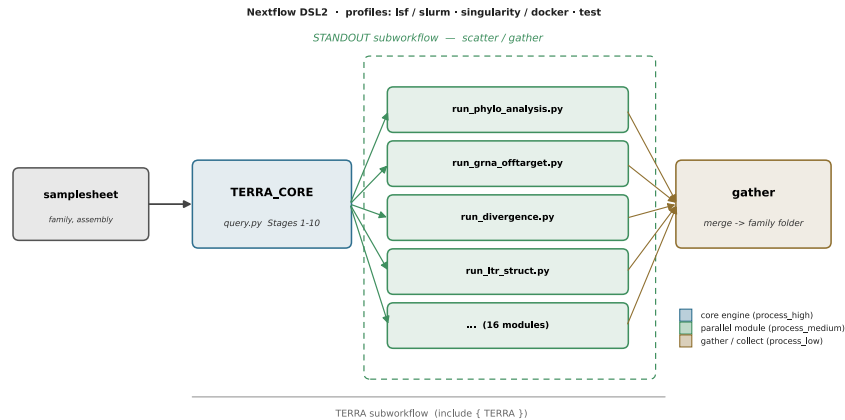


Figure 16: The Nextflow DSL2 execution graph: the core engine feeds a scatter of independent copy-resolved module tasks, gathered into a single per-family output. Profiles select LSF or Slurm and Singularity or Docker.

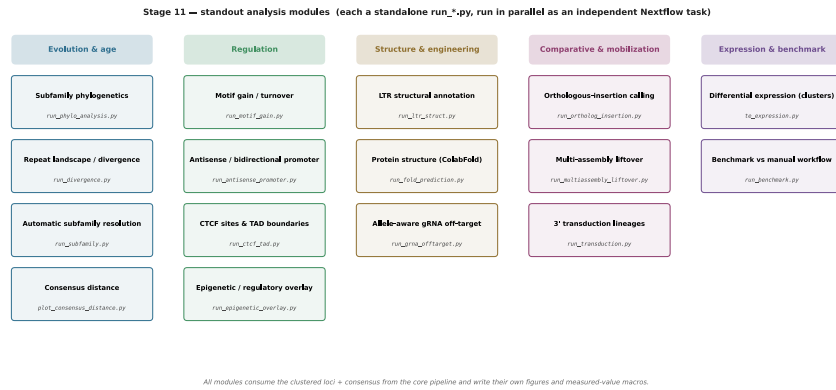


Figure 17: The copy-resolved analysis modules grouped by biological theme. Each card names the standalone `run_*.py` script. The phylogenetics, gRNA, transduction and antisense modules are reported as results in Section 2; the remainder are the additional analysis modules of Section 6.

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